

Secure Vault Application

Create a **frontend-only secure vault application**, similar in concept to **Passbolt** or a **password manager**.

The app should allow users to store secrets securely by **encrypting data before saving it to localStorage**.

Tech Stack

- React or Next.js
- Browser Web Crypto API
- LocalStorage for persistence
- No backend or server code

Functional Requirements

1. Vault & Secrets

Each secret should contain:

- Name
- Username
- Password
- Notes (optional)

Users should be able to:

- Create a secret
- View a list of secrets
- Delete a secret

2. Encryption (Mandatory)

Use the **Web Crypto API**

- Derive an encryption key from a **master password**
- Encrypt secrets **before storing them**

- Store **only encrypted data** in localStorage
- Decrypt secrets **only in memory**

Note: Plaintext secrets must never be stored

3. Lock / Unlock Flow

- Application starts in a **locked state**
- User unlocks vault using the master password
- Vault auto-locks on page refresh

4. Storage

- Use **localStorage** only for:
 - Encrypted vault data
- Master password or encryption key must **not** be persisted

Evaluation Criteria

We will evaluate:

- Correct use of Web Crypto API
- Security awareness (key handling, plaintext avoidance)
- React state management
- Separation of concerns (UI vs logic)
- Error handling and UX clarity

Bonus (Optional)

- Password generator
- Clipboard copy with auto-clear
- Search secrets (decrypt in memory only)
- Clear explanation of security decisions

Constraints (Important)

- No backend / API routes
- No server actions
- No third-party crypto libraries
- No plaintext secret storage

Submission Guidelines

- Share a Git repository link
- Include a short README explaining:
 - Setup instructions
 - Architectural decisions
 - Any assumptions made